

Shell Scripting Practical Questions

Question 1

Q. Display the login name and home directory of the current user.

```
Ques_1.sh

1 #i/bin/bash
2
3 echo "Login name :$USER "
4
5 echo "Home Directory :$HOME "
```

Question 2

Q. Menu-driven program: 1. Date 2. Cal 3. Ls 4. Pwd 5. Exit.

```
Ques_2.sh

1 #i/bin/bash
2
3 choice=1
4 while [ $choice -ne 5 ]
5 do
6     echo ""
7     echo "1.Date"
8     echo "2.Cal"
9     echo "3.Ls"
10    echo "4.Pwd"
11    echo "5.exit"
12    echo "Enter a choice : "
13    read choice
14
15    case $choice in
16        1) date ;;
17        2) cal ;;
18        3) ls ;;
19        4) pwd ;;
20        5) echo "exit"
21    esac
22 done
```

Question 3

Q. Accept a name; if it is a file, show its size; if a directory, list its contents.

```
Ques_3.sh

1 # i/bin/bash
2
3 echo -n "Enter a name : "
4 read str
5
6 if [ -f "$str" ]
7 then
8     ls -s "$str"
9
10 elif [ -d "$str" ]
11 then
12     ls "$str"
13
14 else
15     echo "inavlid data"
16
17 fi
```

Question 4

Q. Check whether a given number is prime.

```
Ques_4.sh

1 # i/bin/bash
2
3 echo "Enter a number : "
4 read n
5
6 res=$(factor $n)
7
8 if [ $(echo $res | wc -w) -eq 2 ]
9 then
10     echo "it is a prime number"
11 else
12     echo "it is not a prime number"
13
14 fi
```

Question 5

Q. Find the greatest of three numbers.

```
Ques_5.sh

1 #!/bin/bash
2
3 echo "Enter three numbers separated by spaces: "
4 read a b c
5
6 if (( a >= b && a >= c ));
7 then
8     echo "The greatest number is: $a"
9 elif (( b >= a && b >= c ));
10 then
11     echo "The greatest number is: $b"
12 else
13     echo "The greatest number is: $c"
14 fi
```

Question 6

Q. Check whether a given year is a leap year.

```
Ques_6.sh

1 # i/bin/bash
2
3 echo "Enter a year : "
4 read year
5
6 if [ `expr $year % 4` -eq 0 ] || [ `expr $year % 100` -eq 0 ]
7 then
8     echo "It is a leap year"
9 else
10     echo "It is not a leap year"
11 fi
```

Question 7

Q. Check whether a number is positive, negative or zero.

```
Ques_7.sh

1 # i/bin/bash
2
3 echo -n "Enter a number : "
4 read N
5
6 if [ `expr $N` -gt 0 ]
7 then
8     echo "Number is positive"
9 elif [ `expr $N` -lt 0 ]
10 then
11     echo "Number is negative"
12 else
13     echo "Number is 0"
14 fi
```

Question 8

Q. Print the multiplication table of a given number.

```
Ques_8.sh

1 # i/bin/bash
2
3 echo -n "Enter a number : "
4 read n
5
6 for var in {1..10}
7 do
8     res=$((n * var))
9     echo "$n x $var = $res"
10 done
```

Question 9

Q. Find the factorial of a given number.

```
Ques_9.sh

1 #i/bin/bash
2
3 echo "Enter a number : "
4 read n
5
6 fact=1
7
8 for ((i=1; i<=n; i++))
9 do
10     fact=$((fact*i))
11
12     echo "Result : $fact "
13 done
```

Question 10

Q. Print the Fibonacci series up to n terms.

```
Ques_10.sh

1 # i/bin/bash
2
3 echo "Enter a number"
4 read n
5
6 a=0
7 b=1
8
9 for (( i=0; i<$n; i++ ))
10 do
11     echo -n "$a"
12
13     h=$(( a+b ))
14
15     a=$b
16     b=$h
17 done
```

Question 11

Q. Calculate gross salary from basic salary (DA 40%, HRA 20%).

```
Ques_11.sh

1 # i/bin/bash
2
3 echo "Enter a basic salary : "
4 read sal
5
6 g_sal=$(echo "$sal + ($sal * 0.4) + ($sal * 0.2)" | bc)
7
8 echo "Gross salary is : $g_sal"
```

Question 12

Q. Check whether a file exists and display its last modified time.

```
Ques_12.sh

1 #i/bin/bash
2
3 f=$1
4 if [ -f $f ]
5 then
6     echo "Last modified time : "
7     ls -l $f | cut -d " " -f9
8     echo "File is exists"
9 else
10    echo "File is not exists"
11 fi
```

Question 13

Q. Display the hidden files in the current directory.

```
Ques_13.sh

1 #i/bin/bash
2
3 echo "Hidden files in directory : "
4 ls -a | grep -Fw "."
```

Question 14

Q. Display the executable files in the current directory.

```
Ques_14.sh

1 #i/bin/bash
2
3 echo "Executable files are : "
4 find -perm -x
```

Question 15

Q. Copy the contents of a source file to a destination file with each line reversed.

```
Ques_15.sh

1 #i/bin/bash
2
3 echo "Enter a source file : "
4 read file1
5
6 echo "Enter a destination file : "
7 read file2
8
9 rev $file1 > $file2
```

Question 16

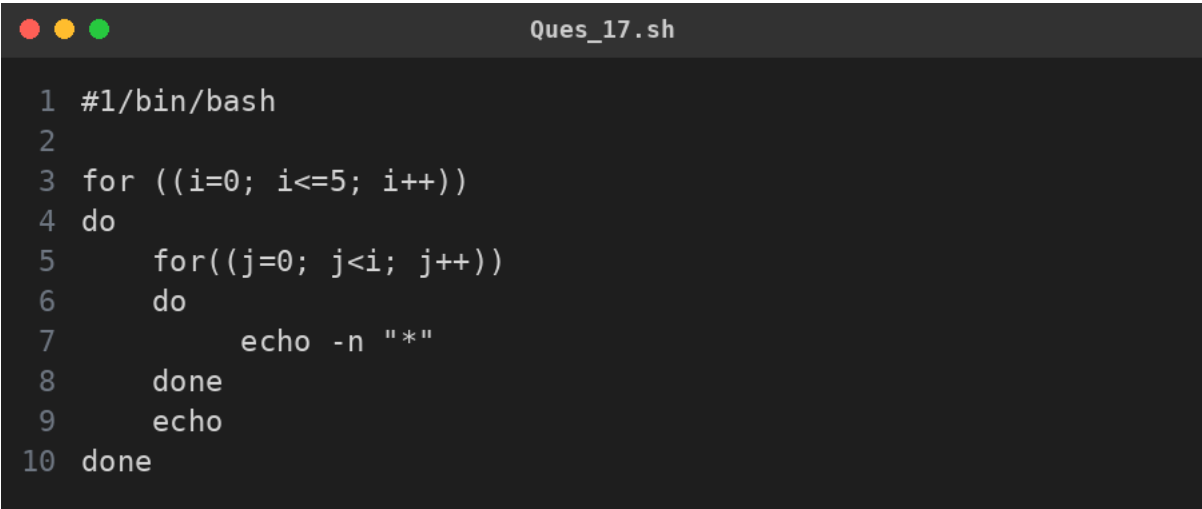
Q. Welcome the user and list the contents of the home directory.

```
Ques_16.sh

1 #i/bin/bash
2
3 echo " Welcome, $USER!"
4
5 echo "Contents of your home directory:"
6 ls -la "$HOME"
```

Question 17

Q. Print a star (*) triangle pattern.



```
1 #!/bin/bash
2
3 for ((i=0; i<=5; i++))
4 do
5     for((j=0; j<i; j++))
6     do
7         echo -n "*"
8     done
9     echo
10 done
```